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RESEARCH ARTICLE

Efficient and Fair Legal Judgment Classification via Personalized GraphRank Summarization and Class-Balanced Learning

K. S. ROSAMMA^{ID}, NAGAMMA PATIL^{ID}, AND C. D. JAIDHAR^{ID}, (Member, IEEE)

Department of Information Technology, National Institute of Technology Karnataka (NITK) Surathkal, Mangaluru, Karnataka 575025, India

Corresponding author: K. S. Rosamma (rosamma.187it501@nitk.edu.in)

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ABSTRACT Legal judgment documents are long, complex, and information-dense, which makes their automatic processing difficult. Transformer-based language models are now widely used in natural language processing because they capture deep contextual relationships, but they struggle with long texts. Their quadratic attention cost and fixed input length often lead to information loss when documents are truncated. This paper presents an efficient and class-imbalance-aware extract-then-classify framework for multi-label legal judgment classification, evaluated on the European Court of Human Rights Article Prediction (ECTHR/A) dataset from LexGLUE. The proposed framework first performs extractive summarization with a Personalized GraphRank (PGR) algorithm that integrates positional priors and TF-IDF centroid priors with Maximal Marginal Relevance (MMR) for redundancy control. The resulting summaries, substantially shorter than the truncated full-text baseline input, retain salient legal content for downstream classification. These summaries are then processed by a DeBERTa-v3-based multi-label classifier trained with class-weighted binary cross-entropy loss and per-label threshold calibration. Experimental results show that the framework achieves performance comparable to a truncated full-text baseline with a substantial reduction in computational cost, while improving macro-F₁ and balanced performance for minority labels under class imbalance. Additional evaluations on token-budget sensitivity, per-label label-wise behavior, proxy intrinsic summary quality, and prior-weight sensitivity further confirm the robustness and practical utility of the proposed framework. Overall, the method provides a transparent, resource-efficient, and class-imbalance-aware approach to legal natural language processing within the present benchmark setting, with improved treatment of imbalanced label distributions.

INDEX TERMS Class-balanced learning, graph-based extractive summarization, legal judgment classification, label imbalance, personalized GraphRank, transformer-based classification.

I. INTRODUCTION

Legal judgments are long and detailed documents. They contain factual information, procedural history, and legal interpretation. Each judgment combines multiple sections that describe events, evidence, and reasoning. Processing these texts is difficult because of their length and complexity. Transformer-based models handle such inputs using a

self-attention mechanism that increases computational cost with longer text. The cost grows quadratically with input length. In real-world legal datasets, many judgments exceed the usual transformer limit of 512–1024 tokens. When this happens, part of the text is truncated, and important information may be lost. This loss reduces the accuracy of legal judgment classification [1].

Another key challenge in legal text classification is label imbalance. Each judgment in the European Court of Human Rights Article Prediction (ECHR/A) dataset [2] is annotated

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with one or more legal articles that represent the rights or violations discussed in the case. This creates a multi-label classification task. In the European Court of Human Rights Article Prediction benchmark within the LexGLUE suite, some articles appear very frequently while others occur only rarely. This uneven distribution lowers macro-level performance and introduces bias in the learning process. As a result, models often favor majority labels and overlook minority categories [3]. Building classification systems that are both efficient and robust under label imbalance is therefore essential for trustworthy legal Natural Language Processing (NLP) [4].

To address these challenges, this study proposes an efficient and class-imbalance-aware extract-then-classify framework. The method begins by selecting concise and representative sentences from each judgment using a Personalized GraphRank (PGR) summarization process. These extracted sentences are then used as input to a transformer-based classifier that operates within a practical token limit. This approach lowers computational cost while retaining salient legal content required for accurate classification.

Traditional graph-based summarization methods such as TextRank [5] and LexRank [6] are interpretable and unsupervised. However, they rely on uniform sentence priors and often generate redundant content, especially in long documents. The proposed PGR improves these methods by extending the PageRank algorithm [7] with two additional priors. The first is a positional prior that applies exponential decay to favor early sentences. The second is a centroid prior based on Term Frequency-Inverse Document Frequency (TF-IDF) similarity [8], which highlights sentences most relevant to the central content of the document. Redundancy is further minimized using Maximal Marginal Relevance (MMR) [9], resulting in concise and diverse summaries.

The extracted summaries are classified using the DeBERTa-v3 base model [10]. This model is fine-tuned for multi-label article prediction. To handle label imbalance, a class-balanced optimization approach is used. It combines weighted binary cross-entropy loss [11] with per-label threshold calibration [12]. This improves recall for minority classes without materially increasing model complexity or inference cost. Together, these methods enhance both efficiency and imbalance-aware performance in legal document classification under class imbalance.

Experiments were performed on the official ECHR/A dataset from LexGLUE to ensure reproducibility. Each model was trained using identical preprocessing, tokenization, and hyperparameter settings. The baseline models include a truncated Full-Text baseline, Lead-K, Random-K, and personalized graph-based extractive variants. Evaluation metrics consist of micro- and macro-F₁ scores [13], Hamming Accuracy [14], Label Ranking Average Precision (LRAP) [15], and Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) [16]. In addition, the evaluation includes token-budget sensitivity analysis, per-label fairness assessment, proxy intrinsic summary evaluation,

and prior-weight sensitivity analysis to provide stronger empirical support for the robustness of the proposed framework. Statistical significance was tested using paired bootstrap confidence intervals and approximate randomization tests.

The main contributions of this work are summarized as follows:

- 1) **PGR:** This is an interpretable and computationally efficient graph-based extraction algorithm. It integrates positional and centroid priors and applies redundancy control to generate concise summaries.
- 2) **Class-Balanced Extractive Classification:** This is an imbalance-aware method under label imbalance that improves macro-F1 and recall for minority labels.
- 3) **Rigorous Evaluation Protocol:** The study follows a standardized experimental setup. It includes reproducible baselines and statistical testing to ensure the reliability of results.
- 4) **Efficiency–Imbalance Analysis:** The framework is evaluated using a quantitative study. The results show how the proposed system balances efficiency, interpretability, and label-balanced performance in legal NLP under class imbalance.

The rest of this paper is organized as follows. Section II reviews related work on graph-based summarization and methods for handling label imbalance in legal text classification. Section III describes the proposed methodology. Section IV presents the experimental setup and results. Section V provides the discussion, and Section VI concludes the paper.

II. RELATED WORK

Research in legal NLP has evolved from rule-based heuristics to deep contextual modeling. Early methods used keyword matching, term frequency, and statistical weighting to extract information from legal texts. These approaches were efficient but lacked contextual understanding. They also failed to capture relations between legal concepts in context. The introduction of distributed word representations such as Word2Vec and GloVe improved text modeling [17]. However, these methods were still inadequate for representing complex legal reasoning and long dependencies. This limitation led to the rise of transformer-based language models, which model contextual interactions more effectively across long text sequences.

In legal summarization, early extractive methods such as TextRank and LexRank used graph-based ranking to identify key sentences. Each sentence was represented as a node, and edges were formed based on lexical similarity. These models were transparent and unsupervised. However, they often produced redundant summaries and showed limited adaptability in long judgments. Later extensions such as PositionRank [18] and TopicRank [19] introduced positional and topical priors to improve diversity and topic coverage. Despite these improvements, they did not explicitly address class imbalance in downstream legal classification settings.

The introduction of transformer models greatly improved legal text processing. Domain-specific pretraining further enhanced this progress. Models such as LegalBERT [20] and CaseLawBERT [21] strengthened contextual understanding for statute interpretation and entailment. Large-scale benchmarks like LexGLUE [2] and LEXTREME [22] established standardized evaluation frameworks. They also enabled consistent comparison across different legal jurisdictions. More recent long-context transformer variants have sought to reduce fixed context limitations through sparse or structured attention mechanisms, but such models still remain computationally demanding for large legal corpora. Despite these advances, transformer-based approaches often remain resource-intensive and may show weaker recall for minority legal categories under label imbalance, creating efficiency, calibration, and interpretability challenges. Sparse-attention and long-context transformer architectures such as Longformer [23] and BigBird-style models [24] are also relevant to long-document legal NLP because they reduce the strict context limitations of standard transformers through structured attention patterns. However, such models generally remain more computationally demanding than lightweight extraction-based classification pipelines, especially when applied to large legal corpora. In the present benchmark-oriented study, the focus is therefore placed on a controlled comparison against a truncated full-text transformer baseline while evaluating whether graph-based extraction can retain competitive performance under substantially lower effective input length.

Recent studies have aimed to balance contextual richness and computational efficiency. Jain et al. [25] introduced DCESumm, a deep clustering and extractive scoring framework that improved the informativeness and diversity of legal summaries. Sharma and Singh [26] proposed RecSumm, a recursive summarization algorithm that outperformed large language models in Recall-Oriented Understudy for Gisting Evaluation (ROUGE) and BLANC scores while preserving factual accuracy. Kalyan et al. [27] applied transformer-based fine-tuning for extractive legal summarization; their results showed good performance but also sensitivity to class imbalance. Dan et al. [28] developed a semantic-structural fusion model using Lawformer and reported substantial ROUGE improvements at a high computational cost. Shreyasi et al. [29] introduced an unsupervised role-labeled knapsack approach (URL-KnapSum), achieving scalability without labeled training data. Sharma and Singh [30] combined InLegalBERT embeddings with K-Means clustering for Indian judgments and obtained strong ROUGE-L scores with moderate interpretability. Sargeant et al. [31] used the Claude 3 Opus large language model for UK case law classification, achieving an F_1 score of 87% while introducing a new legal taxonomy.

More recently, legal NLP has also begun to explore Large Language Models (LLM) and prompt-based approaches for long-text reasoning, structured information extraction, and legal decision support. These directions offer

strong contextual modeling ability, but they also introduce challenges related to computational cost, controllability, and reproducibility in benchmark-oriented settings. Recent prompt-optimization frameworks further suggest that efficient adaptation strategies may provide an alternative direction for improving large-model applications in structured decision-support settings [32]. More broadly, sequence-aware modeling strategies developed for complex long-range dependency patterns in other domains also highlight the potential value of stronger temporal feature extraction mechanisms for structured long-text analysis. In contrast, the present work emphasizes an interpretable extraction-based classification framework for legal judgment prediction with lower computational cost under controlled resource constraints. Recent research, therefore, shows a clear trend toward hybrid systems that combine transformer-based representations with lightweight extraction, ranking, or clustering mechanisms. These approaches aim to retain contextual relevance while reducing computational cost. However, most existing methods still face trade-offs among efficiency, interpretability, and fairness-aware performance under label imbalance.

The review of prior work reveals several persistent research gaps:

- Transformer-based methods often prioritize predictive performance but provide limited interpretability and require substantial computational resources.
- Graph-based and unsupervised models offer computational efficiency but generally do not address label imbalance in downstream multi-label classification.
- Long-context and domain-specific transformer models improve contextual modeling but remain resource-intensive and less transparent.
- Very few frameworks jointly target efficiency, interpretability, and fairness-aware performance under imbalanced legal label distributions.

This study addresses these limitations by integrating graph-based extraction with fairness-aware classification under label imbalance. The proposed method combines positional and centroid priors with class-balanced optimization. This design supports efficient, interpretable, and computationally practical legal judgment classification while improving minority-label performance.

III. METHODOLOGY

This section introduces the methodology used to develop the proposed framework for legal judgment classification. The framework integrates extractive summarization with multi-label classification to achieve efficiency and balanced performance under label imbalance. Figure 1 presents an overview of the proposed framework. The pipeline consists of preprocessing, graph-based extractive summarization, and imbalance-aware multi-label classification. Legal judgments are first segmented into sentences and represented in TF-IDF space. A Personalized GraphRank module then generates a budgeted summary through graph construction,

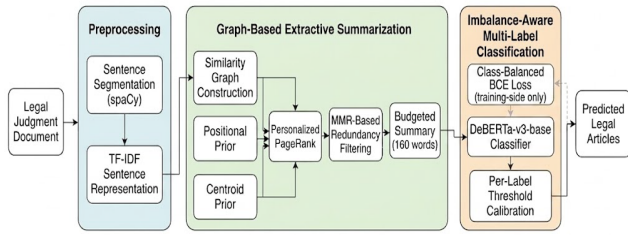


FIGURE 1. Architecture of the proposed efficient and class-imbalance-aware legal NLP framework. Legal judgments are segmented into sentences and represented in TF-IDF space. A graph-based extractive summarization module then applies similarity graph construction, positional and centroid priors, Personalized PageRank, and MMR-based redundancy filtering to generate a budgeted summary. The summary is subsequently classified using DeBERTa-v3-base with class-balanced loss during training and per-label threshold calibration for imbalance-aware multi-label prediction.

prior-guided ranking, and redundancy filtering. The resulting summary is finally processed by a DeBERTa-v3-base classifier with class-balanced optimization and per-label threshold calibration. The following subsections describe these components in detail.

A. DATA PREPROCESSING AND SENTENCE REPRESENTATION

Data preprocessing and sentence representation form the foundation of the proposed framework. Legal judgments are long and complex. They contain factual narratives, procedural details, and statutory references. Effective segmentation and representation are therefore essential to preserve the structure and content of the text before constructing the sentence similarity graph.

Each document is divided into individual sentences using the *spaCy* NLP toolkit [33]. The toolkit applies a combination of rule-based and statistical boundary detection methods. During preprocessing, legal entities, numbers, and citations are retained because they provide context that supports accurate article prediction.

After sentence segmentation, each sentence s_i is represented using a TF-IDF vector. Let $\mathbf{x}_i \in \mathbb{R}^m$ denote the TF-IDF representation of sentence i , where m is the vocabulary size induced by the document collection. These vectors provide a sparse lexical representation suitable for graph construction and centroid-based sentence ranking. The sentence vectors are arranged to form a document matrix, as shown in Equation (1).

$$X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T \in \mathbb{R}^{n \times m} \quad (1)$$

Here, X represents the sentence-document matrix, and n is the total number of sentences in the document. Each row of X corresponds to one sentence vector.

Similarity between sentences is computed using cosine similarity in the TF-IDF space, as defined in Equation (2) [8].

$$\text{Sim}(i, j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (2)$$

In this equation, $\mathbf{x}_i$ and $\mathbf{x}_j$ denote the TF-IDF vectors of sentences i and j , respectively. A higher similarity score $\text{Sim}(i, j)$ indicates stronger lexical relatedness between the sentences. These similarity values are used to form the weighted sentence graph employed in the next stage.

This representation converts each judgment into a compact and interpretable lexical form that supports efficient graph-based extractive summarization before downstream contextual modeling by the transformer classifier.

B. PERSONALIZED GRAPH RANK FOR EXTRACTIVE SUMMARIZATION

After sentence representation, a weighted sentence graph $G = (V, W)$ is constructed. Each vertex in V represents a sentence, and each edge weight in W corresponds to the cosine similarity between a pair of sentence TF-IDF vectors. Only edges with $\text{Sim}(i, j) \geq 0.10$ are retained to suppress weak lexical links and keep the graph sparse.

Sentence importance is estimated using the PGR algorithm. PGR is a personalized variant of PageRank in which sentence salience depends not only on graph centrality but also on document position and centroid relevance.

The first prior, called the *positional prior*, assigns greater importance to sentences appearing earlier in the document. It is defined in Equation (3):

$$\pi_{\text{pos}}(i) = \exp(-0.015 i), \quad (3)$$

where i denotes the sentence index in the document. This exponential decay gives higher weight to earlier sentences, which often contain background facts and procedural context in legal judgments.

The second prior, called the *centroid prior*, measures the closeness of each sentence to the document centroid in the same TF-IDF space. It is defined in Equation (4) [8]:

$$\pi_{\text{cent}}(i) = \cos(\mathbf{x}_i, \mathbf{c}), \quad (4)$$

where $\mathbf{x}_i$ is the TF-IDF vector of sentence i , and $\mathbf{c}$ is the TF-IDF centroid of the document, computed as the mean of the sentence vectors:

$$\mathbf{c} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i. \quad (5)$$

A higher value of $\pi_{\text{cent}}(i)$ indicates that the sentence is more central to the document content.

The two priors are combined to form the personalization vector shown in Equation (6):

$$\mathbf{p} = 0.4 \pi_{\text{pos}} + 0.6 \pi_{\text{cent}}. \quad (6)$$

The coefficients are selected as empirical operating settings for interpretable graph ranking, to balance positional salience and centroid relevance, and their effect is further examined through prior-weight sensitivity analysis in the experimental evaluation. These coefficients are not introduced as theoretically derived or adaptively optimized parameters in the present framework.

Let A denote the thresholded sentence-similarity matrix, and let $\tilde{A}$ be its row-normalized transition matrix. The sentence similarity matrix is symmetric before normalization. The graph is first constructed from undirected similarity scores and then converted into a row-normalized transition matrix for Personalized PageRank. Sentence scores are then computed using the personalized PageRank update rule. [7]

$$\mathbf{r}^{(t+1)} = (1 - \alpha)\mathbf{p} + \alpha\tilde{A}^T \mathbf{r}^{(t)}, \quad (7)$$

where $\mathbf{r}^{(t)}$ is the sentence score vector at iteration t , and $\alpha = 0.85$ is the damping factor. Iteration continues until convergence, that is,

$$\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < 10^{-6}.$$

The resulting vector $\mathbf{r}$ gives the final sentence importance scores.

To reduce redundancy during sentence selection, MMR is applied as shown in Equation (8) [9]

$$s_i = \arg \max_{s_i \in D \setminus S} \left[\lambda \text{Sim}(s_i, \text{Doc}) - (1 - \lambda) \max_{s_j \in S} \text{Sim}(s_i, s_j) \right], \quad (8)$$

where D is the set of candidate sentences and S is the set of already selected sentences. Here, λ controls the trade-off between relevance and novelty. The term $\text{Sim}(s_i, \text{Doc})$ measures how representative a sentence is of the document, while $\text{Sim}(s_i, s_j)$ discourages selecting sentences that are too similar to those already included.

Sentences are selected in descending order of importance until the token budget $B = 160$ is reached. In the experimental evaluation, additional budget settings are also examined to study the sensitivity of downstream classification performance to extraction length. The resulting summary provides a compact extract that retains salient factual and legal content while limiting redundancy.

C. CLASS-BALANCED MULTI-LABEL CLASSIFICATION

The summaries generated by the PGR module are used as input for legal article prediction. Each summary is tokenized to a maximum sequence length of 512 tokens and then encoded using the DeBERTa-v3-base transformer model. DeBERTa employs disentangled attention, which separately models content and positional information. This design supports fine-grained modeling of legal arguments and contextual dependencies in the extracted summaries.

Model training is performed using the AdamW optimizer. The learning rate is set to 2×10^{-5} , the weight decay is fixed at 0.01, and the batch size is 16. A constant random seed value of 13 is used for reproducibility. The model is trained for three epochs, and early stopping based on validation micro F_1 is used to reduce overfitting.

The label distribution in the dataset is highly imbalanced. A few legal articles appear very frequently, while many others are rarely represented. Standard binary cross-entropy may therefore bias learning toward majority labels. To mitigate

this issue, a class-weighted binary cross-entropy loss is used, as defined in Equation (9) [11]:

$$\mathcal{L} = -\frac{1}{C} \sum_{c=1}^C w_c [y_c \log \sigma(z_c) + (1 - y_c) \log(1 - \sigma(z_c))], \quad (9)$$

where C denotes the total number of labels, $y_c \in \{0, 1\}$ is the true label for class c , z_c is the predicted logit, and $\sigma(\cdot)$ is the sigmoid function. The class weight is defined as

$$w_c = \frac{N_{neg,c}}{N_{pos,c}} \quad (10)$$

where $N_{pos,c}$ and $N_{neg,c}$ are the numbers of positive and negative samples for label c , respectively. This weighting increases the contribution of minority labels during optimization.

After training, label-specific thresholds are tuned to balance precision and recall. For each class c , the decision threshold θ_c is selected based on validation performance, as shown in Equation (11) [12]:

$$\theta_c = \arg \max_{\theta \in [0, 1]} F_1^{(c)}(\theta). \quad (11)$$

This per-label threshold calibration adjusts decision boundaries dynamically instead of relying on a fixed global threshold. Its effect is further examined through per-label precision, recall, and F1 analysis to assess label-wise trade-offs under imbalance. It is particularly useful under label imbalance, where different labels exhibit different score distributions.

The combination of class-weighted loss and label-wise threshold tuning improves macro- F_1 and balanced accuracy, thereby supporting imbalance-aware performance under imbalanced label distributions.

D. ALGORITHMS

Algorithms 1 and 2 summarize the complete workflow of the proposed system. The training pipeline integrates sentence representation, graph construction, and Personalized GraphRank extraction, followed by DeBERTa-based multi-label classification with class-balanced optimization and threshold calibration. During inference, the same extraction and calibrated classification process is applied to unseen documents, enabling efficient and imbalance-aware legal article prediction within a fixed token budget.

E. COMPUTATIONAL EFFICIENCY AND REPRODUCIBILITY

The proposed framework achieves computational efficiency through two main strategies: graph sparsification and extractive compression. These steps reduce both processing time and memory requirements without materially affecting downstream classification performance.

Graph construction requires pairwise sentence-similarity computation within a document. Only edges with $\text{Sim}(i, j) \geq 0.10$ are retained, which reduces the number of active graph connections and keeps the ranking stage efficient in practice.

Algorithm 1 Training Pipeline: Personalized GraphRank With Class-Balanced Optimization

-
- 1: **for** each document **do**
 - 2: Segment the document into sentences
 - 3: Compute TF-IDF sentence vectors
 - 4: Build a sentence-similarity graph with $\text{Sim}(i, j) \geq 0.10$
 - 5: Compute π_{pos} and π_{cent} ; combine $\mathbf{p} = 0.4\pi_{pos} + 0.6\pi_{cent}$
 - 6: Run Personalized PageRank ($\alpha = 0.85$) until convergence
 - 7: Apply MMR and select sentences within budget $B = 160$
 - 8: Encode the budgeted summary using DeBERTa-v3-base and compute logits $\mathbf{z}$
 - 9: Update model parameters using class-weighted binary cross-entropy loss
 - 10: **end for**
 - 11: Calibrate label thresholds θ_c on the validation set
-

Algorithm 2 Inference Pipeline: Budgeted Extraction and Calibrated Prediction

-
- 1: Segment the document into sentences
 - 2: Compute TF-IDF sentence vectors
 - 3: Construct the sentence-similarity graph and compute Personalized GraphRank scores
 - 4: Apply MMR and generate an extract under budget $B = 160$
 - 5: Predict label probabilities using DeBERTa-v3-base and output $\hat{Y} = \{c : \hat{y}_c \geq \theta_c\}$ using calibrated label-specific thresholds
-

After graph-based extraction, only the most relevant sentences are selected to form a condensed summary. The extractive compression stage limits the input length for the transformer classifier to $B = 160$ tokens. In transformer architectures, self-attention complexity grows quadratically with sequence length L , that is, $O(L^2)$. Restricting the input from full legal documents to budgeted extracts therefore reduces computational cost substantially and enables efficient processing of long judgments on moderate hardware resources.

Each experiment is repeated three times to verify consistency across runs. The final results are averaged over these runs to reduce random variation. Statistical validation is performed using paired bootstrap confidence intervals and approximate randomization tests. In addition, the evaluation includes proxy intrinsic summary assessment and sensitivity analyses for extraction budget and prior weighting to provide further evidence on robustness and information retention. These procedures support the reliability and reproducibility of the reported findings.

IV. EXPERIMENTAL SETUP AND RESULTS**A. EXPERIMENTAL SETUP**

All experiments were conducted on Google Colab Pro using an NVIDIA A100 GPU with 40 GB of VRAM. The implementation was developed in PyTorch 2.4 with HuggingFace Transformers 4.44 under CUDA 12.2. Mixed precision (bfloat16) and gradient checkpointing were used to reduce memory usage and improve computational efficiency. A fixed random seed value of 13 was used to support reproducibility across runs.

All models used the DeBERTa-v3-base encoder as a common backbone to provide a controlled comparison. The AdamW optimizer was employed with a learning rate of 2×10^{-5} , weight decay of 0.01, and a batch size of 16. Training was limited to three epochs with early stopping based on validation micro- F_1 performance to reduce overfitting. Weighted binary cross-entropy loss compensated for label imbalance, with class weights computed as $w_c = N_{neg,c}/N_{pos,c}$. The token limit was set to 512 for the truncated full-text baseline and 160 for extractive models, maintaining consistent evaluation under computational constraints. In addition to the core system comparison, the evaluation also includes token-budget sensitivity analysis, prior-weight sensitivity analysis, proxy intrinsic summary assessment, and per-label label-wise analysis to examine robustness, information retention, and imbalance-related trade-offs.

To evaluate the effect of different extraction strategies, all systems shared the same classification backbone and optimization setup. The following baselines were selected for comparison:

- **Full-Text:** Processes truncated inputs (512 tokens) to represent a standard transformer-based classification baseline under fixed context length.
- **TextRank:** Serves as a classical unsupervised graph-based baseline for extractive summarization.
- **PGR Extractive:** Extends TextRank using personalized PageRank ($\alpha = 0.85$) with positional and centroid priors ($w_{pos} = 0.4$, $w_{cent} = 0.6$) and MMR-based redundancy control under a 160-token budget.
- **TF-IDF Extractive:** Represents a simple frequency-based scoring approach, providing a low-complexity benchmark.
- **Random-K and Lead-K:** Used as control baselines to measure the effect of random and positional sentence selection.
- **PGR Extractive + Imbalance-Aware Calibration:** Augments the extractive classifier with class-balanced loss and per-label threshold calibration to evaluate improvements under label imbalance.

These baselines were chosen to cover diverse methodological categories—truncated full-text, graph-based, statistical, and control—so that improvements achieved by the proposed framework can be assessed from both practical and statistical perspectives. The additional sensitivity and diagnostic analyses are conducted on top of this controlled setup to evaluate

the effect of extraction length, prior weighting, summary retention, and label-wise calibration behavior.

B. DATASET DESCRIPTION

The evaluation uses the ECTHR/A corpus from the LexGLUE benchmark. The dataset contains 11,000 legal judgments, split into 9,000 training instances, 1,000 validation instances, and 1,000 test instances. Each case is annotated with one or more of 10 legal article labels, forming a multi-label classification task. A statistical overview of the dataset is presented in Table 1.

The dataset exhibits two properties that are central to the present study. First, the label frequency distribution is highly skewed across legal articles, as shown in Fig. 2. The maximum-to-minimum label frequency ratio is 106.06, indicating severe class imbalance. This motivates the use of class-balanced loss and per-label threshold calibration in the classification stage. Second, the documents are long and strongly right-skewed in length, as illustrated in Fig. 3. The average document length is 1,662.08 words, while the median length is 1,032 words. Many judgments therefore extend well beyond the practical input range of standard transformer-based classifiers.

In addition, the dataset has low but non-trivial multi-label density, with an average of 1.163 labels per document. Taken together, these properties make ECTHR/A a suitable benchmark for evaluating both extractive compression and fairness-aware legal text classification under label imbalance. Accordingly, the present empirical findings should be interpreted with respect to this benchmark setting rather than as direct evidence of cross-jurisdictional generalization.

TABLE 1. Summary statistics of the ECTHR/A legal judgment dataset.

Feature	Value
Total documents	11 000
Train / Validation / Test	9 000 / 1 000 / 1 000
Distinct labels	10
Average labels per document	1.163
Average document length (words)	1 662.08
Median document length (words)	1 032
Max/Min label frequency ratio	106.06

C. EVALUATION METRICS

A combination of complementary metrics was used to evaluate prediction accuracy, label-balanced performance under label imbalance, and ranking reliability. Let $Y, \hat{Y} \in \{0, 1\}^{N \times C}$ represent the ground-truth and predicted label matrices, where N is the number of samples and C is the number of labels.

1) MICRO- AND MACRO-F₁

The F_1 score in Equation (12) is the harmonic mean of precision and recall.

$$F_1 = \frac{2TP}{2TP + FP + FN} \quad (12)$$

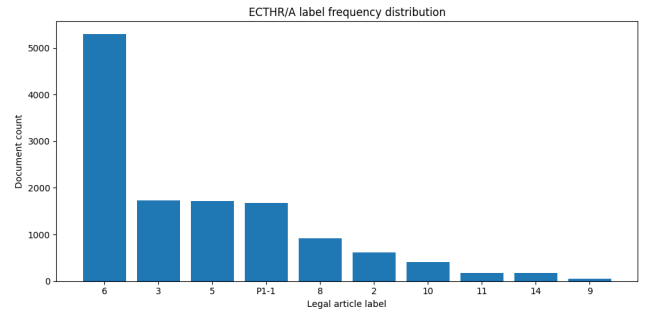


FIGURE 2. Label frequency distribution in the ECTHR/A dataset. The distribution is highly skewed across legal article labels, highlighting the severe class imbalance addressed by the proposed class-balanced classification framework.

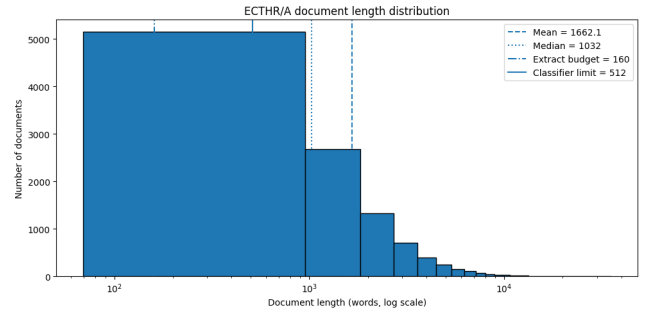


FIGURE 3. Document length distribution in the ECTHR/A dataset on a logarithmic x-axis. The vertical markers indicate the average length, median length, the extractive budget of 160 tokens, and the 512-token limit used in the truncated full-text baseline. The figure shows that many judgments extend well beyond the practical input range of standard transformer-based classifiers.

Here, TP , FP , and FN denote true positives, false positives, and false negatives. Micro- F_1 aggregates counts across all labels, emphasizing common classes. Macro- F_1 computes the average F_1 across labels, giving more weight to minority classes. These two scores together reflect both overall accuracy and label-balanced performance under imbalance.

2) HAMMING AND SUBSET ACCURACY

Hamming Accuracy, defined in Equation (13), measures label-wise correctness.

$$\text{HamAcc} = 1 - \frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C \mathbb{I}[y_{i,c} \neq \hat{y}_{i,c}] \quad (13)$$

The indicator function $\mathbb{I}[\cdot]$ equals 1 when the prediction is incorrect and 0 otherwise. Subset Accuracy, shown in Equation (14), measures the proportion of samples where the full label set is correctly predicted.

$$\text{SubAcc} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[Y_i = \hat{Y}_i] \quad (14)$$

These metrics provide complementary insights—Hamming Accuracy evaluates label-level consistency, while Subset Accuracy measures complete prediction fidelity.

3) BALANCED ACCURACY

Balanced Accuracy computes the average of sensitivity and specificity for each label, and then averages the resulting values across labels. It is defined as

$$\text{BalAcc} = \frac{1}{C} \sum_{c=1}^C \frac{1}{2} \left(\frac{TP_c}{TP_c + FN_c} + \frac{TN_c}{TN_c + FP_c} \right) \quad (15)$$

This metric gives equal importance to positive and negative decisions for each label and is therefore suitable for imbalanced multi-label datasets.

4) LABEL RANKING AVERAGE PRECISION (LRAP)

LRAP evaluates whether true labels are ranked above false labels in the model's probability scores. A higher LRAP indicates better ranking calibration, which is important for legal multi-label tasks where label confidence affects interpretability.

5) ROC-AUC (MICRO AND MACRO)

The Area Under the ROC Curve (AUC) is defined in Equation (16).

$$\text{AUC} = \int_0^1 \text{TPR}(x) d(\text{FPR}(x)) \quad (16)$$

Here, $\text{TPR} = TP/(TP + FN)$ is the true positive rate, and $\text{FPR} = FP/(FP + TN)$ is the false positive rate. Micro-AUC pools all labels together, while Macro-AUC averages them individually to assess label-balanced discrimination. This metric quantifies the model's ability to discriminate between positive and negative instances.

6) STATISTICAL VALIDATION

Each experiment was repeated three times to ensure robustness. Results were validated using 95% bootstrap confidence intervals and approximate randomization tests to support statistical reliability.

Together, these metrics provide a comprehensive evaluation framework. They measure accuracy (Micro/Macro-F1, Hamming), label-balanced performance under label imbalance (Balanced Accuracy and Macro-F1), and ranking reliability (LRAP, ROC-AUC), ensuring an interpretable and reproducible assessment of model performance.

D. RESULTS

Tables 2 and 3 summarize the comparative performance of all systems. In these tables, *PGR Extractive* denotes the proposed personalized graph-based extractive configuration used as the main utility-oriented extractive model, while *PGR Extractive + Imbalance-Aware Calibration* denotes the fairness-enhanced variant with class-balanced loss and per-label threshold calibration. The comparison remains controlled because all systems use the same DeBERTa-v3-base classification backbone and training configuration, while differing only in the input selection strategy.

The strongest utility-oriented performance is achieved by *PGR Extractive*, which attains a micro-F1 of **66.16%** and a Hamming Accuracy of **93.28%**. The truncated *Full-Text* baseline achieves a micro-F1 of 65.72%, a macro-F1 of 47.59%, and the highest LRAP (88.29) and micro ROC-AUC (93.74). Thus, the proposed extractive model remains competitive with the truncated full-text baseline while using a substantially shorter classifier input. In contrast, *TextRank*, *Extractive*, *Random-K*, and *Lead-K* yield consistently lower performance across both accuracy- and ranking-oriented metrics. These results indicate that informed graph-based extraction is more effective than simple unsupervised ranking, frequency-based extraction, or control baselines for preserving legally relevant content.

The imbalance-aware variant, *PGR Extractive + Imbalance-Aware Calibration*, achieves the highest **macro-F1 (56.02%)**, the highest **Balanced Accuracy (82.09%)**, and the highest **macro ROC-AUC (91.29%)** among the evaluated systems. Although its micro-F1 (62.75%) is lower than that of the utility-oriented extractive model, the gain in macro-level performance indicates that class-balanced loss and per-label threshold calibration substantially improve label-balanced discrimination under severe class imbalance. This behavior is consistent with the intended role of the fairness-enhanced configuration, which prioritizes minority-label recognition over purely aggregate micro-level accuracy.

To examine the effect of extraction length, a token-budget sensitivity study was conducted using budgets of 80, 160, 256, and 512 tokens. The results in Table 4 show a monotonic increase in downstream classification performance as the extraction budget grows. At 80 tokens, the model attains a micro-F1 of 60.75% and a macro-F1 of 39.28%. At 160 tokens, performance increases to 66.16% micro-F1 and 45.49% macro-F1. At 256 tokens, the scores further improve to 69.29% and 48.40%, and at 512 tokens to 69.58% and 49.19%, respectively. These findings indicate that larger budgets preserve more predictive content, but they also increase classifier input length and reduce the efficiency benefit of extraction. Therefore, the 160-token setting should be interpreted as a practical efficiency-oriented operating point rather than as the highest-performing budget overall.

A per-label analysis further clarifies the effect of imbalance-aware calibration, as shown in Table 5. The imbalance-aware variant yields the largest F1 gains on low-prevalence labels, including Article 11 (from 0.1500 to 0.6667), Article 14 (from 0.0000 to 0.2609), and Article 9 (from 0.0000 to 0.1622). At the same time, modest declines are observed for some more frequent labels, such as Article 8 (from 0.5314 to 0.5172) and Article 6 (from 0.7059 to 0.6265). This pattern shows that the imbalance-aware configuration improves minority-label recognition while introducing limited trade-offs on certain common labels. The result supports the claim that imbalance-aware calibration improves label-balanced behavior rather than uniformly increasing all performance measures.

TABLE 2. Core performance metrics across comparative systems.

System	Micro-F1	Macro-F1	Ham. Acc.	Sub. Acc.	Bal. Acc.
Full-Text	65.72	47.59	93.21	50.70	70.66
TextRank	64.63	44.35	93.06	50.20	68.85
PGR Extractive	66.16	45.49	93.28	50.20	69.53
Extractive	63.26	42.99	92.95	48.10	68.24
Random-K	62.74	42.34	92.79	46.70	67.81
Lead-K	57.01	39.78	91.84	43.80	66.28
PGR Extractive + Imbalance-Aware Calibration	62.75	56.02	90.86	40.70	82.09

TABLE 3. Ranking and calibration metrics across comparative systems.

System	LRAP	ROC-AUC (micro)	ROC-AUC (macro)
Full-Text	88.29	93.74	87.39
TextRank	86.19	91.57	84.74
PGR Extractive	86.97	92.71	86.82
Extractive	86.47	92.66	87.84
Random-K	86.38	92.24	85.43
Lead-K	80.78	90.13	82.34
PGR Extractive + Imbalance-Aware Calibration	85.07	92.17	91.29

TABLE 4. Token-budget sensitivity analysis.

Budget	Avg. Tokens	Attn. Red.	Micro-F1	Macro-F1	Bal. Acc.	Ham. Acc.	Sub. Acc.	LRAP	ROC-AUC (micro)
80	79.579	37.9136	0.6075	0.3928	0.6616	0.9244	0.460	0.8421	0.9044
160	159.126	9.4822	0.6616	0.4549	0.6953	0.9328	0.502	0.8697	0.9271
256	254.509	3.7067	0.6929	0.4840	0.7125	0.9384	0.531	0.8969	0.9354
512	491.256	0.9949	0.6958	0.4919	0.7185	0.9388	0.539	0.9068	0.9410

TABLE 5. Selected per-label comparison for minority and representative frequent labels.

Label	Prev.	Prec. _{FT}	Rec. _{FT}	F1 _{FT}	Prec. _{Fair}	Rec. _{Fair}	F1 _{Fair}
9	0.005	0.0000	0.0000	0.0000	0.0938	0.6000	0.1622
14	0.016	0.0000	0.0000	0.0000	0.1698	0.5625	0.2609
11	0.037	1.0000	0.0811	0.1500	0.5660	0.8108	0.6667
8	0.123	0.6548	0.4472	0.5314	0.4491	0.6098	0.5172
6	0.299	0.7312	0.6823	0.7059	0.6454	0.6087	0.6265

To assess information retention in the extractive stage, a proxy intrinsic evaluation was performed using the 512-token extract as a longer reference summary. The results of this analysis are presented in Table 6. It shows that intrinsic similarity increases with budget size. For the 80-token budget, the extract achieves ROUGE-1 = 0.2819, ROUGE-L = 0.2775, and BERTScore-F1 = 0.8773 against the 512-token reference. At 160 tokens, these values increase to 0.4097, 0.3172, and 0.8713, respectively. At 256 tokens, the extract reaches ROUGE-1 = 0.6844, ROUGE-L = 0.6773, and BERTScore-F1 = 0.9287. These results indicate that longer extracts preserve more lexical and semantic content, while the 160-token setting retains a meaningful portion of salient legal information under a substantially tighter budget.

The robustness of the Personalized GraphRank design was further examined through prior-weight sensitivity analysis, as shown in Table 7. Four configurations were evaluated: $(w_{pos}, w_{cent}) = (0.2, 0.8), (0.4, 0.6), (0.5, 0.5),$ and $(0.6, 0.4)$. The resulting performance differences are moderate. The original setting $(0.4, 0.6)$ yields 66.16% micro-F1 and 45.49% macro-F1, while nearby settings $(0.5, 0.5)$ and

$(0.6, 0.4)$ achieve slightly higher values of about 66.87% / 46.36% and 66.83% / 46.35%, respectively. The $(0.2, 0.8)$ configuration remains close at 66.27% / 45.40%. These results suggest that the proposed ranking mechanism is reasonably stable across nearby personalization settings and does not depend on a highly fragile single parameter choice within the evaluated benchmark setting. At the same time, this analysis should be interpreted as evidence of local stability rather than as proof of global optimality or adaptive parameter learning.

The efficiency advantage of extraction remains substantial. In the main budgeted setting, the average classifier input is reduced from about 490.0 words for the truncated Full-Text baseline to about 159.1 words for the 160-token extractive configuration. This corresponds to a $3.08\times$ reduction in input length and an approximate $9.48\times$ reduction in self-attention cost under the quadratic complexity assumption. At the same time, the proposed extractive model remains competitive with the truncated full-text baseline on micro-F1, Hamming Accuracy, and ranking reliability. This supports the use of graph-based extraction as an effective efficiency mechanism relative to the truncated full-text baseline evaluated in the present benchmark setting.

V. DISCUSSION

This section discusses the main advantages, trade-offs, and limitations of the proposed framework in light of the experimental results. It also highlights the broader implications of the study and the remaining directions for future work.

TABLE 6. Proxy intrinsic evaluation using the 512-token extract as reference.

Budget	Proxy Ref.	ROUGE-1	ROUGE-L	BERTScore-P	BERTScore-R	BERTScore-F1
80	512	0.2819	0.2775	0.9278	0.8321	0.8773
160	512	0.4097	0.3172	0.8982	0.8462	0.8713
256	512	0.6844	0.6773	0.9534	0.9054	0.9287

TABLE 7. Prior-weight sensitivity analysis.

w_{pos}	w_{cent}	Micro-F1	Macro-F1	Bal. Acc.	Ham. Acc.
0.2	0.8	0.6627	0.4540	0.6948	0.9326
0.4	0.6	0.6616	0.4549	0.6953	0.9328
0.5	0.5	0.6687	0.4636	0.7004	0.9344
0.6	0.4	0.6683	0.4635	0.7008	0.9340

A. COMPARATIVE ADVANTAGES

1) EFFICIENCY AND CONTENT PRESERVATION

Transformer-based classifiers processing longer inputs incur higher computational cost because self-attention grows quadratically with sequence length. In contrast, the proposed framework compresses each judgment to a budgeted extract before classification. Under the main 160-token configuration, the average classifier input is reduced from about 490.0 words for the truncated Full-Text baseline to about 159.1 words, corresponding to a $3.08\times$ reduction in input length and an approximate $9.48\times$ reduction in self-attention cost. At the same time, the *PGR Extractive* configuration remains competitive with the truncated Full-Text baseline on micro-F1, Hamming Accuracy, and ranking-oriented measures. The additional budget-sensitivity analysis further shows that larger extraction budgets improve predictive and intrinsic-retention performance, but they also reduce the computational advantage of extraction. This indicates that the 160-token setting functions as a practical efficiency-oriented operating point rather than the highest-performing budget overall.

2) FAIRNESS UNDER LABEL IMBALANCE

Conventional extractive and transformer-based classifiers often overfit to frequent labels, reducing macro-F1 performance. The proposed fairness-aware model combines class-weight rebalancing with per-label threshold calibration. This configuration achieves the highest macro-F1 (56.02%) and Balanced Accuracy (82.09%) among the evaluated systems. A per-label analysis further shows that the largest gains occur on low-prevalence labels, including Articles 11, 14, and 9, while modest declines remain for some more frequent labels such as Articles 6 and 8. This pattern indicates that the framework improves recognition of minority legal articles while maintaining useful overall predictive performance. This emphasis on balancing minority-label sensitivity with prediction reliability is also consistent with broader high-risk identification settings, where calibration and feature enhancement are evaluated not only by improved detection of rare events but also by their effect on precision and overall operational reliability.

3) TRANSPARENCY AND ACCOUNTABILITY

The graph-based extraction method provides sentence-level evidence for each prediction. Extracted sentences are selected through graph ranking and redundancy-aware filtering, producing an interpretable evidence trail. This traceability improves auditability and interpretability when compared with end-to-end black-box classification over truncated document inputs.

4) ROBUSTNESS OF THE EXTRACTION DESIGN

The additional diagnostic analyses provide further support for the stability of the proposed framework. The proxy intrinsic evaluation shows that lexical and semantic retention improves consistently as the extraction budget increases, while the 160-token setting still preserves a meaningful portion of salient legal content under a much tighter budget. Similarly, the prior-weight sensitivity study shows only moderate variation across nearby personalization settings, indicating that the Personalized GraphRank mechanism does not depend on a highly fragile single weight configuration.

5) COMPUTATIONAL PRACTICALITY

The framework relies on lightweight and general-purpose components such as TF-IDF, personalized PageRank, and DeBERTa-v3-base. This combination supports efficient deployment under moderate computational budgets. Compared with heavier long-context or domain-specific architectures, the proposed framework offers a practical trade-off between performance, interpretability, and resource usage.

6) STATISTICAL ROBUSTNESS

Performance differences were further examined using bootstrap confidence intervals and approximate randomization tests. These statistical checks support the reliability of the reported trends and strengthen the empirical basis of the comparison.

B. LIMITATIONS AND FUTURE SCOPE

Despite strong empirical results, several limitations remain.

1) SENTENCE INDEPENDENCE

The current model treats each sentence as an independent unit during graph construction and ranking. It does not explicitly capture discourse relations, citation-level dependencies, or cross-paragraph interactions. Future work can integrate discourse-aware graph modeling, Graph Neural Networks, or hierarchical transformers to better represent inter-sentence structure.

2) STATIC TOKEN BUDGET

A fixed token budget of 160 tokens is applied as the main operating configuration. Although the added budget-sensitivity analysis shows that larger budgets improve predictive and intrinsic-retention performance, they also reduce the computational advantage of extraction. The present study therefore treats 160 tokens as an efficiency-oriented operating point under controlled constraints rather than as a universally optimal budget for all judgments. Adaptive budgeting based on document length or content density could improve flexibility and extraction quality.

3) DATASET SCOPE AND GENERALIZATION

The evaluation is limited to the ECTHR/A corpus. Although this benchmark is widely used in legal NLP, broader validation on additional legal datasets and jurisdictions would be necessary to establish cross-domain generalization more fully.

4) LIMITED ABSTRACTIVE CAPABILITY

The framework is fully extractive and therefore cannot reformulate or fuse information across sentences. Incorporating structural signals, entity-aware representations, and richer semantic sentence representations may further improve extraction and classification quality in complex judgments.

5) LIMITED STRUCTURAL AND ENTITY-LEVEL MODELING

The current approach does not explicitly model section structure, legal entities, or relations among cited provisions. Incorporating structural signals and entity-aware representations may further improve classification quality in complex judgments.

6) LONG-CONTEXT TRANSFORMER COMPARISON

This study compares the proposed framework against a truncated full-text transformer baseline rather than sparse-attention long-context models. A broader comparison with such architectures would further clarify the trade-off between extraction-based efficiency and long-context representation.

7) LLM-BASED COMPARISON

The present framework is evaluated in a controlled benchmark-oriented setting based on DeBERTa-v3 and graph-based extraction. It does not include direct comparison with LLM- or prompt-based legal NLP pipelines. Such approaches are increasingly relevant for long-text reasoning and structured legal information processing, but they remain outside the scope of the present empirical study.

A further limitation is that early stopping was based on validation micro-F1, which provides a consistent stopping criterion across systems but may not fully reflect macro-level behavior under label imbalance. In addition, the present study does not include evaluation by legal experts, multimodal information, or a separate analysis of highly controversial legal cases, because the work is scoped to benchmark-based

quantitative assessment under a controlled experimental setting. The present study also does not include adaptive or data-driven learning of the prior weights or positional decay factor. These parameters are therefore treated as empirical operating configurations within the current framework rather than as globally optimized or adaptively learned values.

Even with these limitations, the proposed framework provides meaningful operational benefits. It substantially reduces computational cost while maintaining competitive predictive performance relative to the truncated full-text baseline. It also demonstrates measurable improvements in macro-level performance and interpretability under label imbalance. These strengths establish a useful and reproducible baseline for future research on trustworthy and resource-efficient legal NLP systems.

C. IMPLICATIONS

This research supports the broader goal of building sustainable and explainable AI for legal applications. It shows that graph-based extraction, combined with class-balanced calibration, can offer an interpretable and computationally efficient alternative for legal text classification under long-document constraints. Such models are relevant for institutional settings that require accountability, interpretability, and controlled computational cost.

The proposed framework also aligns with emerging AI governance principles, such as those outlined in the EU AI Act (2024) [34]. These principles emphasize transparency, fairness, and human oversight. In this context, the present framework contributes a practical direction for interpretable legal automation, while still requiring broader external validation before deployment across heterogeneous legal environments.

VI. CONCLUSION

This study introduced an efficient and class-imbalance-aware framework for multi-label legal judgment classification. The framework combines PGR summarization with class-balanced optimization to address the challenges of long and imbalanced legal texts. It reduces computational cost through selective extraction while retaining salient legal content required for downstream prediction.

Experimental analysis shows that the proposed framework exhibits two complementary strengths. The utility-oriented *PGR Extractive* configuration achieves a micro-F₁ of 66.16% with substantially reduced classifier input length, while the fairness-enhanced variant achieves the highest macro-F₁ of 56.02% and the highest Balanced Accuracy of 82.09% among the evaluated systems. In the main budgeted setting, the average classifier input is reduced from about 490.0 words for the truncated Full-Text baseline to about 159.1 words, corresponding to a 3.08× reduction in input length and an approximate 9.48× reduction in self-attention cost. These results show that selective extraction can preserve useful contextual information while improving computational efficiency.

The additional evaluation studies further strengthen this conclusion. The token-budget sensitivity analysis shows that larger budgets improve predictive and intrinsic-retention performance, while the 160-token setting remains a practical efficiency-oriented operating point under controlled constraints. The per-label analysis confirms that imbalance-aware calibration yields the strongest gains on minority legal articles, while the proxy intrinsic evaluation indicates that the budgeted extracts retain a meaningful portion of salient legal content. The prior-weight sensitivity analysis further shows that the Personalized GraphRank mechanism remains reasonably stable across nearby personalization settings.

The framework also improves transparency and fairness-aware performance under label imbalance. Sentence-level extraction provides interpretable evidence paths, while class-balanced loss and threshold tuning improve performance on minority labels. By combining graph-based personalization with imbalance-aware optimization, the system achieves a practical balance among efficiency, interpretability, and label-balanced performance. Overall, the proposed PGR-based pipeline presents a benchmark-grounded, interpretable, and resource-efficient alternative for legal text classification under long-document constraints. It provides a foundation for scalable legal analytics and interpretable decision-support settings while meeting practical computational requirements.

Future extensions of the present framework include adaptive budgeting for documents of varying complexity, broader validation across additional legal datasets and jurisdictions, richer discourse- or structure-aware modeling for more advanced legal reasoning settings, and direct comparison with long-context or LLM-based legal NLP approaches.

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K. S. ROSAMMA is currently pursuing the Ph.D. degree in information technology with the National Institute of Technology Karnataka (NITK) Surathkal. She has over eight years of academic experience and has served as an Assistant Professor at various higher education institutions. Her research interests include machine learning and deep learning for natural language processing.



NAGAMMA PATIL received the Ph.D. degree in computer science and engineering from Indian Institute of Technology Roorkee, India. She is currently an Associate Professor with the Department of Information Technology, National Institute of Technology Karnataka (NITK) Surathkal. She has supervised multiple doctoral dissertations and has published extensively in SCI- and Scopus-indexed journals. Her research interests include data mining, soft computing, big data analytics, machine learning, and bioinformatics. She has received awards for her papers and has successfully led funded projects under the Vision Group on Science and Technology, Government of Karnataka.



C. D. JAIDHAR (Member, IEEE) received the Ph.D. degree in computer science and engineering from the National Institute of Technology (NIT) Trichy, India. He is currently a Professor with the Department of Information Technology, National Institute of Technology Karnataka (NITK) Surathkal. He has guided several doctoral research scholars and authored numerous articles in high-impact journals. His research interests include computer networks, cybersecurity, the Internet of Things, and precision agriculture.

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